



Project Delivery

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Large-Scale Software Architecture

2026-I

1. Objective

The objective of this project delivery is to design, model, and verify the architecture of the **National Emergency Response and Crisis Management System**, a System of Systems (SoS) described in the project specifications. Starting from the SoS description, teams must carry out the full architectural engineering process: define the architecture, formalize it as a metamodel and textual models, identify quality attribute concerns, and verify how well the architecture addresses them through an iterative simulation-based process.

The delivery is organized in groups of five teams. Each group jointly owns the global SoS architecture, and each team within the group is responsible for one of the five constituent subsystems throughout all activities.

2. Activities

2.1. Metamodel

All five teams in the group must collaboratively define a single **textX grammar** that serves as the metamodel for the SoS architecture. The metamodel must be expressive enough to represent the architecture at two levels:

- **SoS level:** the five constituent systems, the inter-system connectors, and the information flows between them.
- **Subsystem level:** the internal components (software, hardware, and physical), the internal connectors, and the external interfaces that link each subsystem to the SoS.

The metamodel must also capture **quality-oriented architectural elements** introduced to address quality attribute concerns (see Section 2.3). This includes new component types, connector types, and

attributes that express quality-related properties (e.g., **reliabilityLevel**, **maxResponseTime**, **encryptionType**).

The final grammar file must be identical across all five teams' submissions.

2.2. Architectural Models

Using the metamodel defined in Section 2.1, the group and each team must produce two textual architectural models.

SoS-Level Model (Group Work)

A shared textual model representing the global SoS: the five constituent systems, their inter-system connectors, and the information exchanges between them as described in the project specifications. This model must reflect all quality-driven modifications introduced to address SoS-level quality concerns. It must be identical across all five teams' submissions.

Subsystem Model (Team-Specific Work)

Each team must produce a textual model of its assigned subsystem, written in accordance with the shared metamodel. The model must include:

- The SoS-level integration view: how the subsystem connects to the overall architecture through its external interfaces.
- The full internal C&C structure: all software components, internal connectors, and quality-oriented elements introduced to address the subsystem's quality attribute concerns.

2.3. Quality Attribute Scenarios

Global Operational Scenario (Group Work)

The group must define a single **global operational scenario** that represents the SoS operating as a whole under real or extreme emergency conditions, capturing how the five subsystems interact dynamically to deliver a coordinated response. This scenario is the operational context from which quality attribute concerns emerge at the SoS level.

For each identified quality attribute, the group must define a **quality scenario** using the following structure:

Element	Description
Stimulus	The event that activates the scenario.
Source	The origin of the stimulus (external system, user, sensor, network, attacker).
Environment	The operational conditions when the event occurs (peak load, degraded connectivity, active emergency).
Artifact	The architectural element(s) involved (systems, subsystems, components, connectors).
Response	The expected SoS behavior or reaction.

Element	Description
Response Measure	The measurable criterion used to evaluate whether the response is satisfactory.

Subsystem-Level Quality Scenarios (Team-Specific Work)

Each team must define **at least three quality scenarios** for its assigned subsystem, focusing on the primary quality attribute challenge described for that subsystem in the project specifications. Each scenario must follow the same six-element structure and be traceable to specific elements of the subsystem's architectural model.

2.4. Verification Process

Each team must verify the architecture of its assigned subsystem through at least **four iterative verification cycles**, using the architectural models and quality scenarios defined above. The verification must be implemented in a **Jupyter Notebook** (Python recommended, Google Colab preferred), following the modeling and simulation approach developed in Laboratory 5.

Each iteration must include:

- Instantiation of the architectural model according to the metamodel.
- Graphical representation of the architecture (C&C and deployment view (if applicable)).
- Simulation of a quality attribute scenario.
- Analysis and interpretation of results.

The four iterations must follow this progression:

- **Iteration 1:** Verify one quality attribute scenario on the initial subsystem architecture.
- **Iteration 2:** Introduce architectural tactics or patterns that improve the quality attribute analyzed in Iteration 1. Re-execute the verification and evaluate the improvement.
- **Iteration 3:** Verify a different quality attribute scenario on the evolved architecture.
- **Iteration 4:** Introduce additional tactics or patterns to address the scenario from Iteration 3. Verify their effect.

Each team must also synchronize its results with the global SoS model, providing evidence that local quality improvements contribute to the overall SoS-level quality objectives. A **summary table or comparative chart** must close the notebook, showing how the architecture evolved across iterations and how quality attributes improved as a result.

3. Delivery

3.1. Artifact

The artifact consists of the following elements:

1. **Metamodel (metamodel.tx):** Final shared textX grammar capturing both SoS-level and subsystem-level architectural concepts, including quality-oriented elements. Collaborative file, identical across all group submissions.

2. **SoS Model (`sos_model.sos`)**: Shared textual model of the global SoS consistent with the metamodel. Collaborative file, identical across all group submissions.
3. **Subsystem Model (`subsystem_model.sos`)**: Textual model of the team's assigned subsystem, consistent with the metamodel, including the full internal C&C structure and quality-oriented elements.
4. **Verification Notebook (`verification.ipynb`)**: Jupyter Notebook containing group and team information, system name and description, the quality scenarios defined for the subsystem, executable code for all four verification iterations with textual explanations and result analysis, evidence of SoS-level verification synchronization, and a closing summary of architectural evolution and quality improvement.
5. **Report (`report.pdf`)**: A document describing the global operational scenario, the SoS-level quality scenarios, the subsystem-level quality scenarios with traceability to model elements, the verification methodology, the tactics and patterns applied across iterations, and the conclusions.

3.2. Submission Format

The deliverable must be submitted via Google Classroom in a single `.zip` file named **group-G-team-X-p**, where G = group number (1, 2) and X = team letter (1a, 2b, ..., 2a, 2b, ...). The `.zip` file must contain:

- `metamodel.tx`
- `sos_model.sos`
- `subsystem_model.sos`
- `verification.ipynb`
- `report.pdf`

Note: Only one team member must submit the file.

3.3. Delivery Deadline

Monday, May 4, 2026, 23h59.

3.4. Project Presentation

Tuesday, May 5, 2026, 18h00 - 20h00.